Synthetic Handbook — Cable Sizing and Earthing (DEMO-CB-003 Rev C)

SYNTHETIC DEMONSTRATION DOCUMENT — fictional data, not a real standard

Fictitious handbook demonstrating current-carrying capacity tables, derating factors and short-circuit sizing. All coefficients are invented for the demo corpus and rounded as stated.

Chapter 1 — Base ratings for XLPE cables

Base table (single circuit buried directly, depth 0.7 m, soil resistivity 1.0 K·m/W, ambient earth 20 °C, copper conductor, 600/1000 V LV or 8.7/15 kV MV XLPE):

Cross-section	LV 4-core in ground	MV 1-core in ground	LV clipped direct, 30 °C
16 mm²	105 A	95 A	85 A
70 mm²	220 A	200 A	164 A
240 mm²	464 A	400 A	—
300 mm²	522 A	444 A	—

Round derived currents to 1 A as shown in the worked examples below.

Chapter 2 — Derating factors (synthetic)

Condition	Factor
Ambient earth 25 °C	0.96
Ambient earth 30 °C	0.91
Soil resistivity 2.5 K·m/W	0.71
Grouping: 2 circuits	0.90
Grouping: 3 circuits	0.83
Protective conduit in earth	0.80

2.1 Worked example (authoritative for verification tests)

A 240 mm² LV XLPE cable is buried in one trench of three circuits at ambient earth 30 °C and soil resistivity 2.5 K·m/W. The current-carrying capacity is I = 464 × 0.91 × 0.71 × 0.83 = 254 A after rounding to 1 A. A 16 mm² cable under the same conditions carries 105 × 0.91 × 0.71 × 0.83 = 57 A.

Group factor alone for two circuits reduces the 240 mm² rating to 464 × 0.90 = 417 A.

2.2 Short-circuit sizing

Minimum cross-section for a prospective fault: S = I·√t / k with k = 143 for copper XLPE 90 °C. Example: 25 kA for 3 s requires S = 25000 × √3 / 143 = 302 mm² (rounded to 1 mm²). Therefore a 240 mm² cable is NOT adequate for a 25 kA 3 s fault; it is adequate up to 22.2 kA for 3 s.

Chapter 3 — Earthing requirements

Item	Requirement	Notes
Substation earth grid	≤ 1.0 Ω	measured, summer condition
LV panel main earth terminal	≤ 5 Ω	soil dependent
Motor body earth	≤ 0.5 Ω	bonding check
Minimum earth conductor	25 mm² copper	static and protective
Earthing lead to grid	≥ 50 mm² Cu equivalent	fault-rated
Temporary protective earth	240 mm² set	for 11 kV works
Grid conductor corrosion allowance	300 mm² steel	100 years design

The 600/1000 V LV XLPE cable shall be used where the nominal phase-to-earth system voltage exceeds 600 V only for test leads, which is prohibited in permanent wiring.

3.1 Temperature note

Ampacity factors are valid for ambient earth 20 to 30 °C only; at 35 °C apply an additional factor 0.86.